ON THE RIGIDITY OF THE MINIMUM-MODULUS PATTERN: A CIRCLE-ROTATION READING

CLAUDE

ABSTRACT. The minimum moduli $\mu_n^{(c)}$ of a family of characteristic polynomials attached to $\tau(p_n)$ oscillate, and the locations of their extrema obey a strikingly rigid arithmetic pattern. This note argues that the rigidity is not special to τ: granted a single dominant complex frequency, it is the universal rigidity of the return times of an irrational circle rotation (the three-gap theorem), and the specific pattern is a two-term continued fraction made visible. The genuinely open questions are isolated at the end. The note is self-contained; the two analytic inputs it borrows are stated as facts in §2 and proved in a companion note.

1. SETUP AND NOTATION

Let τ be the Ramanujan tau function, defined by $\Delta(q) = q\prod_{k\geq 1}(1-q^k)^{24} = \sum_{m\geq 1}\tau(m)\,q^m$, and let p_n denote the n-th prime. Fix the sequence $h_0 = 1$, $h_n = \tau(p_n)$ for $n \geq 1$.

The recurrence and its data. Equation (D) of the manuscript,

$$n\,h_n = \sum_{r=1}^{n} j_r\,h_{n-r} \qquad (n \geq 1),$$

is a bijection between sequences $(h_n)_{n\geq 0}$ with $h_0 = 1$ and *data sequences* $(j_r)_{r\geq 1}$. In generating-function form, with $H(t) = \sum_{n\geq 0} h_n t^n$ and $J(t) = \sum_{r\geq 1} j_r t^r$, it reads $tH' = HJ$, i.e. $J = t\,(\log H)'$.

The deformation. For a parameter c, prepend c to the data sequence, $j^{(c)} = (c, j_1, j_2, \dots)$, and run (D) forward from $h_0 = 1$ to obtain the deformed sequence $h^{(c)} = (h_n^{(c)})_{n\geq 0}$. Equivalently the generating function is multiplied by an exponential,

$$H^{(c)}(t) = e^{ct}\,G(t), \qquad G(t) = \exp\int_0^t J(s)\,ds,$$

so that $h_n^{(c)} = [t^n]e^{ct}G(t)$. Throughout we take $c = 1$, the case in the computational run.

The spectra. To each n the manuscript attaches the monic degree-n polynomial (Theorem 2.3)

$$\chi_n^{(c)}(x) = \sum_{r=0}^{n} \binom{n}{r} r!\, h_r^{(c)}\, (-1)^r\, x^{\,n-r},$$

and writes $\mu_n^{(c)} = \min\{\, |\zeta| : \chi_n^{(c)}(\zeta) = 0 \,\}$ for the modulus of its root nearest the origin. It is the sequence $(\mu_n^{(c)})_n$, plotted against n, whose oscillation is at issue.

2. The two analytic inputs

The argument below borrows exactly two facts. Both are established in the companion asymptotics note; here they are simply named, with the constants defined as they arise.

Input 1 (dominant singularity). Since $J = tH'/H$, the singularities of G are the zeros of $H(t) = 1 + \sum_{n\geq 1} \tau(p_n)t^n$. The zero nearest the origin is a complex-conjugate pair $t_0, \bar{t}_0$; writing it in polar form

$$t_0 = \rho\, e^{i\theta}, \qquad \rho = |t_0| \approx 0.0435, \qquad \theta = \arg t_0 \approx 0.2432\,\pi,$$

this defines the modulus ρ and the *argument* θ. (Numerically t_0 is stable under truncation of H, and ρ is corroborated by the coefficient growth $|[t^n]G|^{1/n} \to 1/\rho$.)

Input 2 (coefficient form and the minima). By singularity analysis [12, 13], a dominant conjugate pair at $t_0 = \rho e^{i\theta}$ forces the leading-order shape

$$h_n^{(c)} \;\sim\; C\,\rho^{-n}\,\cos\!\big(n\theta + \varphi\big),$$

in which $C > 0$ is a real constant and φ is a fixed *phase offset* (it absorbs the argument of the complex amplitude attached to the singularity and the shift contributed by the e^{ct} factor; only its value, not its origin, matters below). Because $\chi_n^{(c)}$ is monic, the product of its roots is $n!\,h_n^{(c)}$, and with the bulk of the spectrum varying smoothly in n the variation of $\mu_n^{(c)}$ tracks $|h_n^{(c)}|$. Consequently $\mu_n^{(c)}$ dips toward zero exactly at the sign changes of $h_n^{(c)}$, i.e. where the phase

$$n\theta + \varphi \quad \text{is nearest an odd multiple of} \quad \tfrac{\pi}{2}.$$

3. The empirical regularity to be explained

A run to $n = 500$, followed by a difference-table analysis of the lower and upper envelopes of $(\mu_n^{(1)})_n$, reports the following. The successive local minima occur at indices whose consecutive gaps are drawn from exactly *two* values, $\{4, 5\}$; the gap 5 occurs on a strict period-9 schedule; and the second and third differences of the gap sequence are perfectly periodic with period 9 (no jitter anywhere). One full block is

$$8 \times 4 + 1 \times 5 = 37,$$

containing 9 minima, so the pattern recurs with period 37 in n; the upper envelope is the same, phase-shifted. The average gap is $37/9 = 4.111\ldots$, and an independent Fourier analysis of the same data places its dominant peak at period 4.10. So a discrete method (difference tables) and a spectral method (FFT) agree on a fast period ≈ 4.1 and a long period 37.

4. The strictness is the fingerprint of a circle rotation

A sinusoid sampled at the integers would not produce gaps from exactly two values with the 5 on a strict period-9 schedule and perfectly periodic higher differences; it would jitter. The pattern of §3 — two gap lengths, one inserted quasi-periodically — is the signature of a *Sturmian* (mechanical) word [1, 2], and Sturmian words are exactly the symbolic codings of an irrational rotation of the circle.

Input 2 makes the rotation explicit. The minima of $\mu_n^{(c)}$ fall where the phase $n\theta+\varphi$ is nearest an odd multiple of $\pi/2$; tracking them is tracking the returns of the orbit $\{\, n\,(\theta/\pi) + \beta\,\}$ near $\frac{1}{2}$ on the circle, an equidistributed orbit in the sense of Weyl [9, 10]. The *three-gap theorem* — conjectured by Steinhaus, first proved by Sós, Surányi, and Świerczkowski [3, 4, 5, 6] — states that the gaps between successive returns of any irrational rotation take at most three distinct values, arranged quasi-periodically. Here they take two, $\{4, 5\}$. So the rigidity is not a special fact about τ: it is the universal rigidity of rotation return times. Once one grants the single dominant frequency θ of Input 1, the strict two-value pattern is *forced*; nothing else was available.

5. The specific pattern is the continued fraction made visible

The average gap is $37/9 = 4.111\ldots$, and as a continued fraction

$$\frac{37}{9} = [4; 9] = 4 + \frac{1}{9}.$$

That two-term expression *is* the observed law: the integer part 4 is the base gap; the partial quotient 9 is the period of the correction (one extra step — a 5 — every nine). The minima locations are, to this order, a Beatty sequence $\lfloor k\,(37/9) + \beta \rfloor$ [7, 8]. The fast period π/θ predicted by Input 2 equals $1/0.2432 \approx 4.11$, matching 37/9. So the puzzle of why an $8 \times 4 + 5$ block repeats with period 37 collapses to a two-term continued fraction; the Fourier peak at 4.10 is the blurred analytic shadow of this crisp arithmetic object, which is why the difference-table view looks far more rigid than a spectral peak has any right to.

6. Where the interesting number theory actually lives

Two facts remain unexplained by the model.

First, that $H(t) = 1 + \sum_n \tau(p_n)t^n$ has a single, clean, well-isolated dominant complex zero at all (Input 1). That one conjugate pair should govern everything is a genuine statement about the analytic structure of a *prime-indexed* τ series — an object with no modularity, no functional equation, and no Euler product. The dominant-singularity principle [12, 13] says what such a zero *would* control, not why this series should have so clean a one. I know no reason.

Second, and more pointed: the value of the rotation number θ/π and its proximity to 9/37. Is it *exactly* 9/37 or merely very close? This is decidable, and the meaning hinges on it.

- If θ/π is *irrational* with continued fraction $[0; 4, 9, a_3, \dots]$ and a_3 large, then 9/37 is an excellent rational approximant, the coding is Sturmian [1, 2], and near $n \sim 37\, a_3$ a *second-order defect* must appear — a block with seven or nine 4's rather than eight, or two consecutive 5's. The run to $n = 500$ shows about thirteen clean cycles with no such defect, which already forces a_3 to be large: $|\theta/\pi - 9/37| \lesssim 1/(37 \cdot 500) \approx 5 \times 10^{-5}$.
- If θ/π were *exactly* 9/37, the pattern would be exactly period-37 forever. That would be astonishing: it would place the nearest zero of a prime-indexed τ series at a rational multiple of π, demanding an arithmetic explanation.

My honest bet is the first — irrational, $[0; 4, 9, a_3, \dots]$ with a_3 large — because there is no reason a transcendental-looking θ assembled from τ at primes should be a rational multiple of π, and every real number masquerades as a small-denominator rational when known to only four digits. I hold it loosely; the second case, if true, would be worth a paper.

7. How to settle it

Pin θ down to eight or ten digits (rather than the four currently in hand), form θ/π, and read its continued fraction [8]. If it begins $[0; 4, 9, a_3, \dots]$, then *a_3 predicts the exact n at which the first second-order defect must occur* — a prediction tested by extending the run past $n = 500$ to that point. A defect at the predicted place proves the rotation-number account and shows θ/π irrational; no defect to all computable precision would be the surprising case. In the language of circle dynamics this is just reading the rotation number and its convergent denominators off the orbit [11].

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